

WIN

COMPETITIVE

BEHIND

tok/s

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ours / llama / ratio

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PPL = quality (lower better)

MODEL	PREFILL pp512	DECODE tg128	PPL
Qwen3.6-35B-A3B — CROWN (MoE+DeltaNet) the model the engine was built for — beats llama on BOTH ► At the front. Bespoke int-dot Q8 MoE kernel (E_ffn=512, top-2). llama backend: SYCL master	+7.6% 1144 / 1064	+3.5% 84.1 / 81.31	6.45
Qwen3.6-27B — DeltaNet hybrid hybrid DeltaNet + dense MLP ► Prefill wins big (oneDNN GEMM); decode at parity. llama backend: Vulkan	1.9× 577 / 303	parity 10.0 / 9.72	5.34
Qwen3 / Qwen2 dense (8B) dense family baseline (Mistral / Llama / Phi / distills inherit this) ► Prefill wins; DECODE behind — the Q6_K GEMV cliff (Lever 2). llama backend: SYCL	+14.9% 1190 / 1036	0.56× 43.7 / 77.7	2.94
Qwen3-Coder-30B-A3B — qwen3moe standard top-8 MoE (E_ffn=768) ► PREFILL 8.1× (GPU router + generalized int-dot down) — was 0.10×. llama backend: Vulkan	0.66× 651 / 984	0.64× 37.4 / 58.6	11.98

ours / llama

Wave-1 validated (correctness; ride dense path):
Mistral-Small-3.2-24B (PPL 7.42) · DeepSeek-R1-Distill-Qwen/Llama · Phi-4 (PPL 8.25). Perf inherits the dense profile: prefill competitive, decode behind (Lever 2).

Multi-GPU 72B (tensor-parallel):
decode 10.4 tok/s = 1.44× over our own layer-split (NOT a llama A/B) · PPL 8.97.

FOCUS NEXT — two transferable kernel investments that turn broad coverage into broad competitive perf

✓ MoE-expert GEMM kernel — DONE (8.1×)

GPU-gemm router (host loop was 66% of prefill wall) + generalized int-dot W4A8 down for E_ffn≠512 (new moe_qwen3.cpp; crown moe_fused.cpp untouched).
→ qwen3moe prefill 80.6→651 (12×→1.5× behind); MoE shapes now Next-80B-ready

2 Dense-decode GEMV

The Q6_K cliff / repacked / split-K (the P3b known-hard). Closes the dense-family decode loss (43.7 vs 77.7) shared by Mistral / Llama / Phi / distills.
→ turns the whole validated dense family competitive on decode

⚠ Backend-fairness caveat

Crown + dense use llama SYCL (llama's strongest Intel backend); the 27B + qwen3moe rows use llama Vulkan (it has the GPU DeltaNet shaders / was on hand). A llama SYCL re-run is pending for the fairest qwen3moe gap— treat that ratio as approximate, not exact.

Same physical B70, model VRAM-resident for both engines. Every figure traces to docs/benchmark_matrix_2026-06-09.md · no GPU run for this chart.